

The background features a dark blue gradient with faint, light blue concentric circles and degree markings (40, 150, 160, 170, 180, 190, 200, 210, 220, 230, 240, 250, 260) on the left side, suggesting a circular or orbital theme.

Database System Labs - Administration

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OBJECTIVES

- Authorization
- Backup/Restore DB

AUTHORIZATION

Forms of authorization to modify the database schema

- **Index** - allows creation and deletion of indices.
- **Resources** - allows creation of new relations.
- **Alteration** - allows addition or deletion of attributes in a relation.
- **Drop** - allows deletion of relations.

AUTHORIZATION 授予 SPECIFICATION IN SQL

- The **grant** statement is used to confer authorization
grant <privilege list>
on <relation name or view name> **to** <user list>
- <user list> is:
 - a user-id
 - **public**, which allows all valid users the privilege granted
 - A role (more on this later)

AUTHORIZATION SPECIFICATION IN SQL (CONT.,)

- **Granting** a privilege on a view does not imply granting any privileges on the underlying relations.
- The grantor of the privilege must already hold the privilege on the specified item (or be the database administrator).

PRIVILEGES IN SQL

- **select**: allows read access to relation, or the ability to query using the view
 - Example: grant users U_1 , U_2 , and U_3 **select** authorization on the *instructor* relation:
grant select on instructor to U_1 , U_2 , U_3
- **insert**: the ability to insert tuples
- **update**: the ability to update using the SQL update statement
- **delete**: the ability to delete tuples.
- **all privileges**: used as a short form for all the allowable privileges

REVOKING 收回 AUTHORIZATION IN SQL

- The **revoke** statement is used to revoke authorization.

revoke <privilege list>

on <relation name or view name> **from** <user list>

- Example:

revoke select on *branch* from U_1, U_2, U_3

ROLES角色

- **create role** instructor;
- **grant *instructor* to Amit;**
- Privileges can be granted to roles:
 - **grant select on *takes* to *instructor*;**
- Roles can be granted to users, as well as to other roles
 - **create role *teaching_assistant***
 - **grant *teaching_assistant* to *instructor*;**
 - *Instructor* inherits all privileges of *teaching_assistant*

OTHER AUTHORIZATION FEATURES

- **references** privilege to create foreign key
 - **grant reference** (*dept_name*) **on** *department* **to** Mariano;
 - why is this required?
- transfer of privileges
 - **grant select on** *department* **to** Amit **with grant option**;
 - **revoke select on** *department* **from** Amit, Satoshi **cascade**;
 - **revoke select on** *department* **from** Amit, Satoshi **restrict**;

DATABASE BACKUP/RESTORE

- 数据
- 二进制日志
- 代码
- 服务器配置文件
- ...

DATABASE BACKUP/RESTORE

- 增量备份、差异备份、完全备份
- 冷备份、热备份、温备份
- 物理备份、逻辑备份

DATABASE BACKUP/RESTORE

- 物理备份

- 对InnoDB数据进行备份: `mysqlbackup`
- 对MyISAM数据进行备份: `mysqlhotcopy`
- 也可使用文件系统级别的`cp`、`scp`、`tar`、`rsync`等命令

DATABASE BACKUP/RESTORE

- 逻辑备份

- 逻辑备份所有的数据库

`mysqldump -u root -p --all-databases --lock-all-tables > dbbak.sql`

- 逻辑备份指定的数据库xxxDB1、xxxDB2

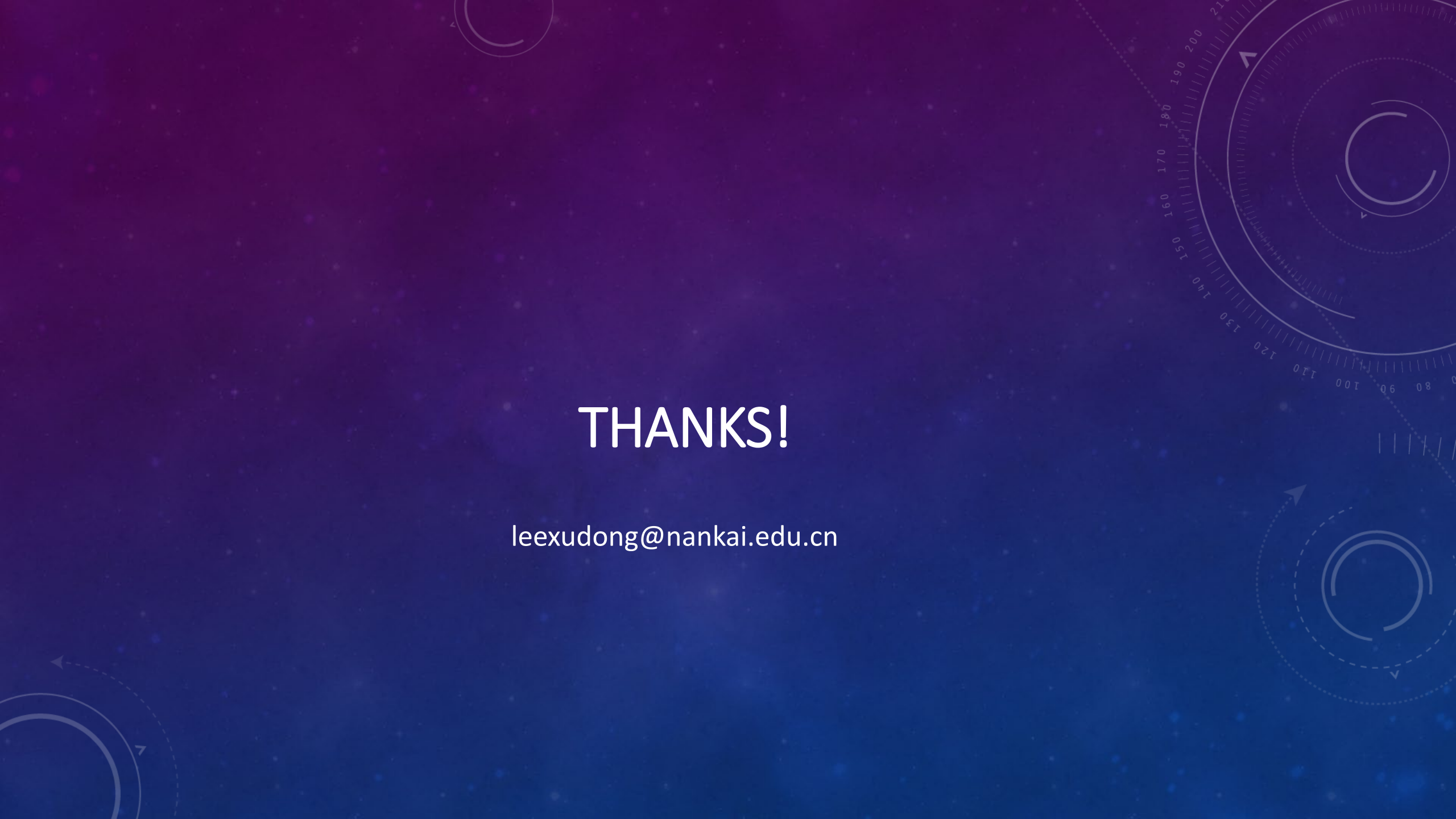
- `mysqldump -u root -p --databases xxxDB1 xxxDB2 > dbbak.sql`

- 恢复指定的数据库:

- `mysql -u root -p xxxDB1 < dbbak.sql`

QUIZ





THANKS!

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